# SMART INDIA HACKATHON 2026

**Problem Statement ID -** 26231

**Problem Statement Title -**Digital Companion for Field Drug Testing

**Theme -** MedTech / BioTech / HealthTech

**PS Category -** Software

**Team ID -** K26234

**Team Name -** HexTech

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# PROPOSED SOLUTION

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## Problem

India had 1,20,010 NDPS cases in 2023 (NCRB). Each needs a field test.

- Officers mainly judge the kit's colour change by looking at it, so lighting and personal judgement can affect the result.
- There is no standard colour value to compare the test result with.
- The test does not always leave a clear digital record of who performed it, where, and when, making it harder to verify later.

## Solution

- Check the Test - The mobile app captures the colour change and helps check the result.
- Record the Test - Saves the result along with the time, location, and Inspector ID.
- Manage the Records - The admin dashboard manages Inspectors, test records, evidence, and reports.

## User Experience

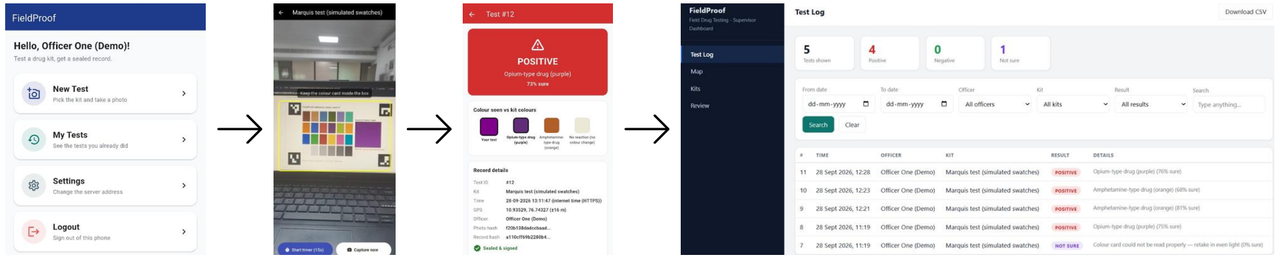
- **4 steps** - choose kit, take photo inside the box, see result, it gets saved. One test takes around 16 seconds (we checked in 19 scans).
- **Bad photo?** - app says NOT SURE, tells the reason (blurry, shiny or card not seen) and asks to take the photo again.
- **Easy to read** - result comes in big letters with colour, in simple English. Next we will try it with 10 officers.

**Prototype Demo Video:** Prototype Video Link

## Our Approach

Test Kit → Take Photo → Check Colour → Get Result → Save Proof → Admin Review

**Prototype Flow -** real screenshots of our working prototype




*Test Log (right side) can search and filter by date, officer, kit and result, and download as CSV or PDF with QR.*

## Novelty

Eye check is guesswork, lab is slow, **FieldProof gives result on the spot for ₹10 with proof**

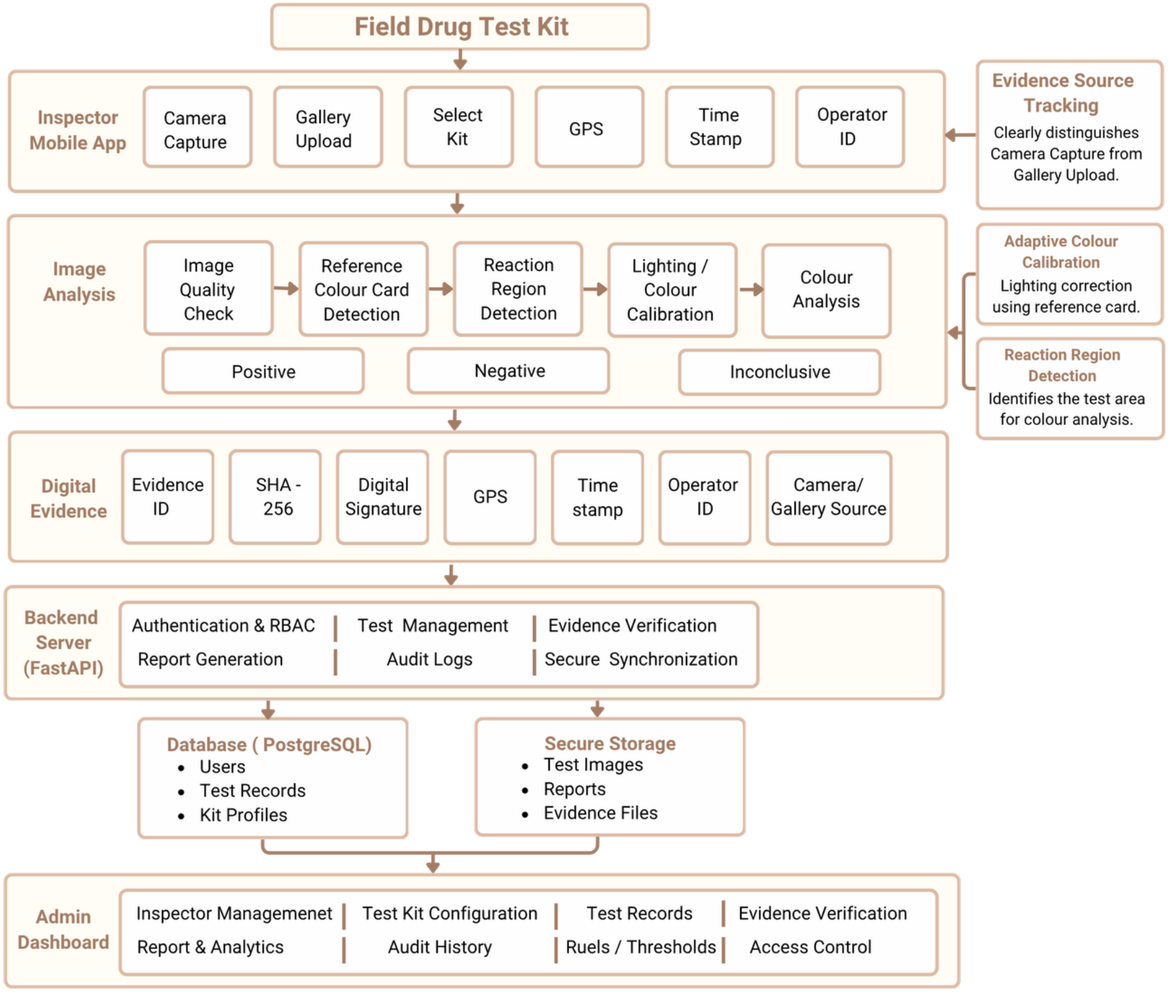
- **Handles Lighting Variations** - Uses a low-cost colour card to improve colour detection under different lighting.
- **Handles Unclear Results** - Blurry, shiny, or unclear images are marked INCONCLUSIVE instead of giving a wrong result.
- **Creates Verifiable Evidence** - Securely records the photo, GPS, time, and Inspector ID, with QR-based verification.
- **Supports Multiple Kits** - New kit colour profiles can be added from the Admin website without changing the app.

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# TECHNICAL APPROACH

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## System Architecture



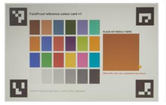

## Our Goals - Measured Results

- Testing: 37 computer made photos and 25 real phone scans.
- Accuracy: all 25 computer made photos correct. Real scans had 0 wrong answers (14 correct, others said NOT SURE).
- Colour Check: Used $\Delta E \leq 14$, 4-point colour matching, card-fit and glare thresholds.
- Speed: Result secured in 2.5 s typically, 5 s maximum.
- Tamper Test: All 4/4 modified records were detected successfully.

*Kit colours are simulated swatches (test photos + card on a screen), real kits need an FSL calibration trial.*
*Right now it is a lab prototype (TRL 4) tested on 1 phone. Next we test real kits with FSL on more phones.*

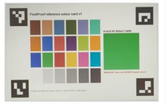
### Proof from Real Phone Scans (29 Sep 2026)


Purple card
**POSITIVE 79%**


Orange card
**POSITIVE 76%**


Plain card
**NEGATIVE 75%**


In-between card
**NOT SURE**


Green card (not in kit)
**NOT SURE**


Blurry photo
**NOT SURE, retake**

We scanned using our app on 1 phone, in 1 room with window light. Card was shown on a laptop screen. We checked each card by eye, not using the app.

**All 19 real scans**

| Card vs app | POSITIVE | NEGATIVE | NOT SURE |
|---|---|---|---|
| Positive card (8) | 5 | 0 | 3 |
| Plain card (4) | 0 | 4 | 0 |
| Trick card (7) | 0 | 0 | 7 |

**0 wrong answers**

All 9 clear scans correct
All 7 trick cards said NOT SURE
3 blurry photos, app asked to retake
Around 16 seconds per test (1 team member)

## Technology Stack

- Flutter (Dart)
- FastAPI (Python)
- OpenCV + NumPy
- CIELAB + ΔE2000
- PostgreSQL + SQLAlchemy
- React + Vite
- JWT + bcrypt
- SHA-256
- Ed25519
- NTP
- Leaflet + OpenStreetMap
- ReportLab

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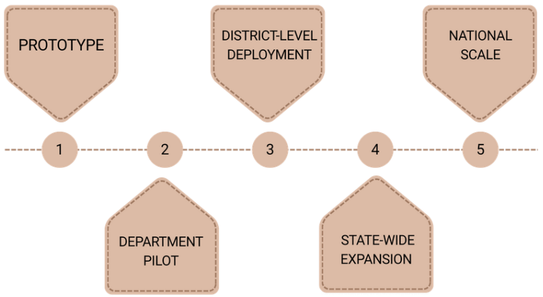
# FEASIBILITY AND VIABILITY

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## Feasibility & Sustainability

- **Technical** - Core app tested with OpenCV-based colour analysis.
- **User-Friendly** - Capture one photo and get a result within seconds.
- **Low Cost** - Uses an existing Android phone, a reusable ~₹10 colour card, and open-source tools.
- **Operational** - Simple photo-based workflow with minimal training required.
- **Scalable** - Add new test-kit colour profiles through the Admin website without updating the app.
- **Sustainable** - Government-cloud hosting, regular profile updates, and secure data storage support long-term use.

## Phased Implementation & Scalability




### Pilot Plan (proposed)

- **Where** - 1 district in Tamil Nadu, with local police and the nearest FSL
- **Who** - 20 officers, for 3 months
- **Check** - every field result is checked again in the FSL lab
- **Success** - 95% or more same as lab, NOT SURE below 10%, every record passes the check
- **Then** - first district, then state, then whole country (phases on the left)
- **Cost** - around ₹10,000 for 3 months, for cards and server (rough estimate)

## Potential Challenges & Strategies

| Real-world problem | What our app does |
|---|---|
| Bad light (street lamp / torch) | Colour card fixes the colours |
| Shiny or blurry photo, card missing | Says NOT SURE and asks to retake |
| Colour is in between two results | Says NOT SURE; senior officer checks |
| Fake photo or edited record | In-app camera only + photo fingerprint |
| Different kits and phones | Kit settings + colours fixed every photo |

## Cost & Deployment Plan

**After pilot,** the state police or NCB IT budget can pay for it.
It is open source with no licence fee, so running cost is low.

| COST | HOSTING | MAINTENANCE | DEPLOYMENT | SCALE |
|---|---|---|---|---|
| Uses an existing phone, ₹10 reusable colour card, and open-source software. | Runs on government cloud for the app, database, and admin dashboard. | Regular security updates and easy kit-profile updates as needed. | Start with a pilot, test in the field, then expand deployment. | New test kits can be added through the website without updating the app. |

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# IMPACT AND BENEFITS

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## Inspector Benefits

- Faster, more consistent testing
- Less manual colour checking
- INCONCLUSIVE when unclear
- Easy digital test records
- Quick access to past test records

## Organisation Benefits

- Proof of every test: time, place, officer & photo
- Secure and verifiable records
- Less paperwork with PDF & QR
- Easy search, reports and monitoring
- Better field-test documentation

## QR-Based Record Verification

**FieldProof - Field Test Record**

Presumptive field result — not a substitute for laboratory confirmatory testing.

**POSITIVE**
Opium-type drug (purple) | 76% sure



Scan to check this record
http://127.0.0.1:8000/v/11

| | |
|---|---|
| Test number | 11 |
| Test ID | 4878cf1d-8f00-4f66-9f73-30c50ed39e62 |
| Saved at (UTC) | 2026-09-28T06:58:23.601Z (time from: internet time (HTTPS)) |
| Phone time | 2026-09-28T12:28:21+05:30 (difference: -2.6 s) |
| GPS location | 10.935299, 76.743282 (±14.791999816894531 m) |
| Officer | Officer One (Demo) (user id 1) |
| Device ID | android-1e42a9c170701656ac8a22b0ca44a6e0 |
| Kit | Marquis test (simulated swatches) [DEMO-MARQUIS-SIM], profile version 1 |
| Lab sample ID | - |
| Review status | NOT_NEEDED |

**FieldProof · Record check**

**✓ Record is genuine**
The photo and details were not changed after they were saved.

**What we checked**

- ✓ The photo is the same as when it was taken
- ✓ The saved details match their fingerprint (hash)
- ✓ The FieldProof server's digital signature is valid
- ✓ The record is correctly linked to the record before it
- ✓ The result, officer and kit shown are the ones that were signed

**Record details**

| | |
|---|---|
| Test number | 11 |
| Result | POSITIVE |
| What the kit showed | Opium-type drug (purple) |
| How sure | 76% |
| Kit | Marquis test (simulated swatches) (version 1) |
| Officer | Officer One (Demo) |
| Saved at (UTC) | 2026-09-28T06:58:23.601Z (internet time (HTTPS)) |
| GPS | 10.93530, 76.74328 |
| Test ID | 4878cf1d-8f00-4f66-9f73-30c50ed39e62 |
| Photo fingerprint (SHA-256) | 294b35454c7e963c7e0f939b4f0e9cd10a279e69fca62634d61ea030d98a9e57 |

## Map - Geo-Tagged test result




## Tamper Test - we tried it on our saved records

**Test #10**
prev 2ece5c80
hash 6b1b4e7f

→

**Test #11**
prev 6b1b4e7f
hash 03dd9f0c

→

**Test #12**
prev 03dd9f0c
hash a110cff6

Ed25519 signature of test 11
JRIdu6ERgadC+2n6chKfUStp...
Photo SHA-256
294b35454c7e963c7e0f939b...

| What we changed (test #11) | Check page | Failed check |
|---|---|---|
| Nothing (original) | GENUINE | none, all 5 passed |
| 1 byte of the photo | TAMPERED | photo |
| Result inside sealed copy | TAMPERED | hash, signature, fields |
| Result in database row | TAMPERED | fields |
| The signature | TAMPERED | signature |

We tried this on our real saved records on 29 Sep 2026, only in memory. Each test keeps the hash of the test before it.

## Future Scope

- Offline Mode - Test even without internet and sync data later.
- Phone Security - Use the phone's security chip to sign each test.
- Police System Integration - our CSV export already has time, GPS, officer, result, photo hash and verify link. Linking with e-Sakshya and CCTNS needs NIC or MHA approval.
- Lab Comparison - Compare field results with lab results to improve accuracy.
- Regional Languages - Support Hindi and regional languages with voice assistance.

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# RESEARCH AND REFERENCES

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| What they found | How we use it | Research |
|---|---|---|
| **IEEE EMBC 2016**<br>Phone camera colours change with light and phone model; reference colours can help handle this. | Our ₹10 colour card fixes colours in every test | IEEE EMBC 2016 – Illumination and device independence for colorimetric detection of urinary biomarkers with smartphone |
| **IEEE TPS-ISA 2024**<br>Digital evidence must maintain integrity and show how it was handled. | Photo fingerprint + signature + chained log | IEEE TPS-ISA 2024 – Digital Evidence Chain of Custody: Navigating New Realities of Digital Forensics |
| **Forensic Science International 2014**<br>A mobile phone camera can measure drug-test colours using RGB values. | Our app reads the test colour as a number | Forensic Science International 2014 – Real-time quantitative colorimetric test for methamphetamine detection using digital and mobile phone technology |
| **Analytical Chemistry 2016**<br>By-eye colour interpretation is subjective; smartphone image analysis can provide objective analysis. | App gives POSITIVE / NEGATIVE / NOT SURE | Analytical Chemistry 2016 – Objective Method for Presumptive Field-Testing of Illicit Drug Possession Using Centrifugal Microdevices and Smartphone Analysis |
| **Digital Discovery 2023**<br>Computer vision can quantify and compare colour changes that may look similar by eye. | App shows photo colour next to kit colours | Digital Discovery 2023 – Teaching old presumptive tests new digital tricks with computer vision |